

HUNDRED DAYS OF CODE

DAY 19

QUESTION:

The algorithm for calculating the Semester Grade Point Average (SGPA) and Cumulative Grade Point Average

(CGPA) for 'm' semesters is given below. Each semester has 'n' courses, where 'n' may vary between semesters.

Assume that the student has passed all courses. Prepare a test-case table covering valid, invalid, and boundary

inputs; write the C program based on the given algorithm; compile and execute it; and test it against the prepared

test cases.

Algorithm

1. Start.
2. Define the variables and their types: global_id, m, n, semester, course, code, credit, GP, semester_TC, semester_TGC, cumulative_TC, and cumulative_TGC as integers; grade as a character; and SGPA and CGPA as floating-point variables.
3. Get global_id.
4. Validate global_id: If global_id is not positive, display an appropriate error message and request it again. Repeat until a positive value is entered.
5. Get m.
6. Validate m: If m is not positive, display an appropriate error message and request it again. Repeat until a positive value is entered.
7. Set the cumulative values: cumulative_TC = 0, cumulative_TGC = 0
8. Repeat Steps 8.1 to 8.8 for m semesters:
 - 8.1. Set the semester totals:
semester_TC = 0, semester_TGC = 0
 - 8.2. Get n.

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8.3. Validate n: If n is not positive, display an appropriate error message and request it again. Repeat until a positive

value is entered.

8.4. Repeat Steps 8.4.1 to 8.4.9 for n courses:

8.4.1. Get code.

8.4.2. Validate code: If it is not positive, display an error message and request it again. Repeat until a positive

value is entered.

8.4.3. Get credit.

8.4.4. Validate credit: If it is not positive, display an error message and request it again. Repeat until a positive

value is entered.

8.4.5. Get grade.

8.4.6. Validate grade: Accept only O, A, B, C, D, E, or P. If the grade is invalid, display an error message and

request it again. Repeat until a valid grade is entered.

8.4.7. Find GP corresponding to grade using the university's conversion scale:

O = 10, A = 9, B = 8, C = 7, D = 6, E = 5, P = 4

8.4.8. Calculate the semester total grade credits:

$\text{semester_TGC} = \text{semester_TGC} + (\text{GP} * \text{credit})$

8.4.9. Calculate the semester total credits:

$\text{semester_TC} = \text{semester_TC} + \text{credit}$

8.5. Calculate the SGPA of the current semester:

$\text{SGPA} = \text{semester_TGC} / \text{semester_TC}$

8.6. Display semester and SGPA.

8.7. Add the semester grade credits to the cumulative grade credits:

$\text{cumulative_TGC} = \text{cumulative_TGC} + \text{semester_TGC}$

8.8. Add the semester credits to the cumulative credits:

$\text{cumulative_TC} = \text{cumulative_TC} + \text{semester_TC}$

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9. Calculate the CGPA:

$CGPA = \text{cumulative_TGC} / \text{cumulative_TC}$

10. Display global_id, m, cumulative_TC, and CGPA.

11. Stop

PSEUDOCODE:

BEGIN

DECLARE global_id, m, n

DECLARE semester, course, code, credit, GP

DECLARE semester_TC, semester_TGC

DECLARE cumulative_TC, cumulative_TGC

DECLARE grade

DECLARE SGPA, CGPA

INPUT global_id

WHILE global_id <= 0 DO

 PRINT "Invalid Global ID. Enter a positive value:"

 INPUT global_id

END WHILE

INPUT m

WHILE m <= 0 DO

 PRINT "Invalid number of semesters. Enter a positive value:"

 INPUT m

END WHILE

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cumulative_TC $\leftarrow$ 0

cumulative_TGC $\leftarrow$ 0

FOR semester $\leftarrow$ 1 TO m DO

semester_TC $\leftarrow$ 0

semester_TGC $\leftarrow$ 0

INPUT n

WHILE n $\leq$ 0 DO

PRINT "Invalid number of courses. Enter a positive value:"

INPUT n

END WHILE

FOR course $\leftarrow$ 1 TO n DO

INPUT code

WHILE code $\leq$ 0 DO

PRINT "Invalid course code. Enter a positive value:"

INPUT code

END WHILE

INPUT credit

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```
WHILE credit <= 0 DO
    PRINT "Invalid credit. Enter a positive value:"
    INPUT credit
END WHILE
```

```
INPUT grade
```

```
WHILE grade is not O, A, B, C, D, E or P DO
    PRINT "Invalid grade. Enter a valid grade:"
    INPUT grade
END WHILE
```

```
IF grade = 'O' THEN
    GP ← 10
ELSE IF grade = 'A' THEN
    GP ← 9
ELSE IF grade = 'B' THEN
    GP ← 8
ELSE IF grade = 'C' THEN
    GP ← 7
ELSE IF grade = 'D' THEN
    GP ← 6
ELSE IF grade = 'E' THEN
    GP ← 5
ELSE
    GP ← 4
END IF
```

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$\text{semester_TGC} \leftarrow \text{semester_TGC} + (\text{GP} \times \text{credit})$

$\text{semester_TC} \leftarrow \text{semester_TC} + \text{credit}$

END FOR

$\text{SGPA} \leftarrow \text{semester_TGC} / \text{semester_TC}$

PRINT "Semester =", semester

PRINT "SGPA =", SGPA

$\text{cumulative_TGC} \leftarrow \text{cumulative_TGC} + \text{semester_TGC}$

$\text{cumulative_TC} \leftarrow \text{cumulative_TC} + \text{semester_TC}$

END FOR

$\text{CGPA} \leftarrow \text{cumulative_TGC} / \text{cumulative_TC}$

PRINT "Global ID =", global_id

PRINT "Number of Semesters =", m

PRINT "Cumulative Total Credits =", cumulative_TC

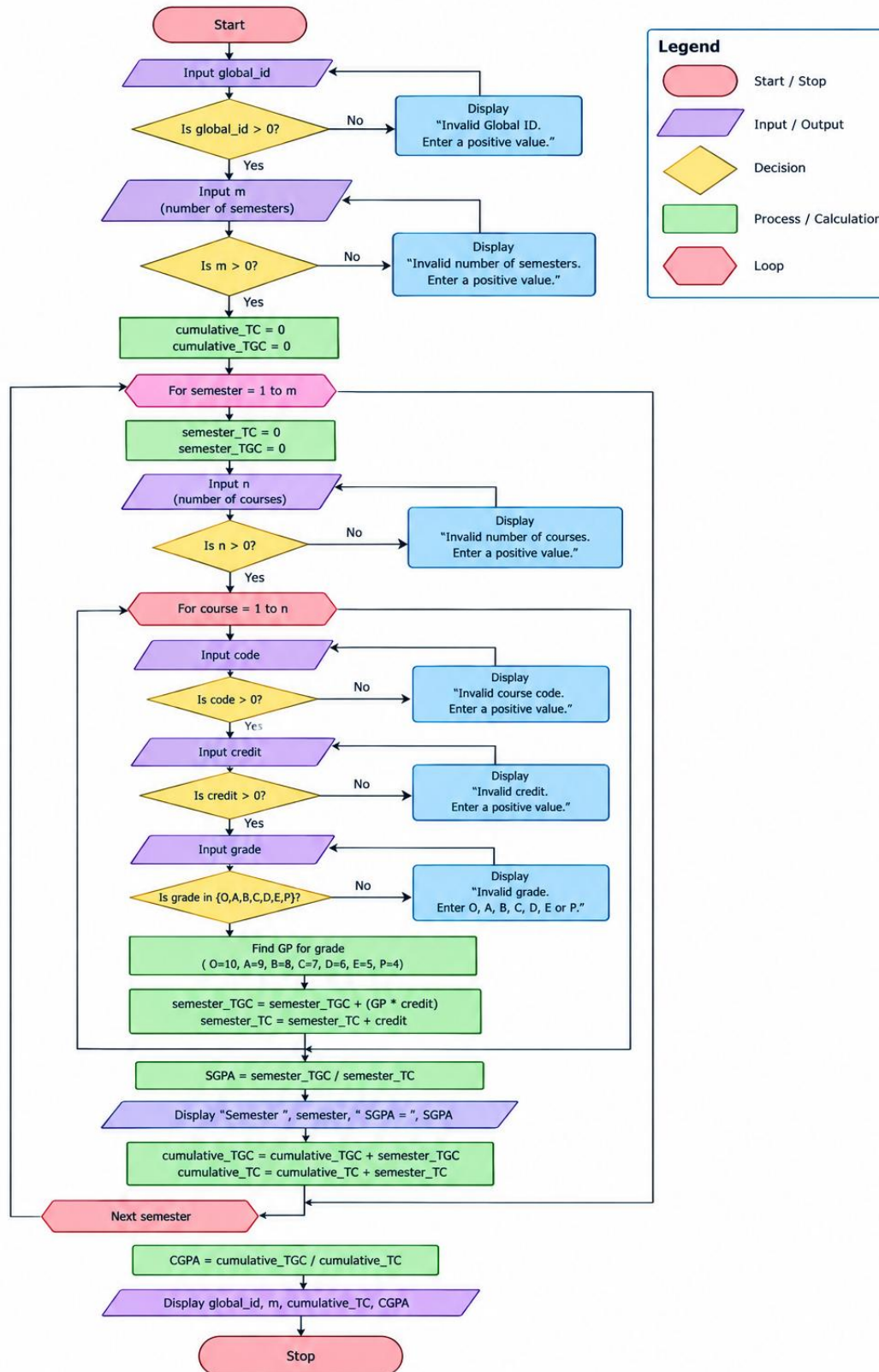
PRINT "CGPA =", CGPA

END

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FLOWCHART:



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C PROGRAM:

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int global_id, m, n;
```

```
    int semester, course, code, credit, GP;
```

```
    int semester_TC, semester_TGC;
```

```
    int cumulative_TC, cumulative_TGC;
```

```
    char grade;
```

```
    float SGPA, CGPA;
```

```
    /* Input and validation of Global ID */
```

```
    printf("Enter Global ID: ");
```

```
    scanf("%d", &global_id);
```

```
    while (global_id <= 0)
```

```
    {
```

```
        printf("Invalid Global ID. Enter a positive value: ");
```

```
        scanf("%d", &global_id);
```

```
    }
```

```
    /* Input and validation of number of semesters */
```

```
    printf("Enter number of semesters: ");
```

```
    scanf("%d", &m);
```

```
    while (m <= 0)
```

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```
{  
    printf("Invalid number of semesters. Enter a positive value: ");  
    scanf("%d", &m);  
}  
  
/* Initialize cumulative values */  
cumulative_TC = 0;  
cumulative_TGC = 0;  
  
/* Process each semester */  
for (semester = 1; semester <= m; semester++)  
{  
    semester_TC = 0;  
    semester_TGC = 0;  
  
    /* Input and validation of number of courses */  
    printf("\nEnter number of courses for Semester %d: ", semester);  
    scanf("%d", &n);  
  
    while (n <= 0)  
    {  
        printf("Invalid number of courses. Enter a positive value: ");  
        scanf("%d", &n);  
    }  
  
    /* Process each course */  
    for (course = 1; course <= n; course++)
```

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```
{  
    printf("\nCourse %d\n", course);  
  
    /* Course code */  
    printf("Enter course code: ");  
    scanf("%d", &code);  
  
    while (code <= 0)  
    {  
        printf("Invalid course code. Enter a positive value: ");  
        scanf("%d", &code);  
    }  
  
    /* Credit */  
    printf("Enter credit: ");  
    scanf("%d", &credit);  
  
    while (credit <= 0)  
    {  
        printf("Invalid credit. Enter a positive value: ");  
        scanf("%d", &credit);  
    }  
  
    /* Grade */  
    printf("Enter grade (O/A/B/C/D/E/P): ");  
    scanf(" %c", &grade);
```

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```
while (grade != 'O' && grade != 'A' &&
      grade != 'B' && grade != 'C' &&
      grade != 'D' && grade != 'E' &&
      grade != 'P')
{
    printf("Invalid grade. Enter O, A, B, C, D, E or P: ");
    scanf(" %c", &grade);
}

/* Grade point conversion */
switch (grade)
{
    case 'O':
        GP = 10;
        break;

    case 'A':
        GP = 9;
        break;

    case 'B':
        GP = 8;
        break;

    case 'C':
        GP = 7;
        break;
```

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```
case 'D':  
    GP = 6;  
    break;  
  
case 'E':  
    GP = 5;  
    break;  
  
case 'P':  
    GP = 4;  
    break;  
}  
  
/* Calculate semester totals */  
semester_TGC = semester_TGC + (GP * credit);  
semester_TC = semester_TC + credit;  
}  
  
/* Calculate SGPA */  
SGPA = (float)semester_TGC / semester_TC;  
  
printf("\nSemester %d SGPA = %.2f\n", semester, SGPA);  
  
/* Update cumulative totals */  
cumulative_TGC = cumulative_TGC + semester_TGC;  
cumulative_TC = cumulative_TC + semester_TC;
```

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```
}

/* Calculate CGPA */

CGPA = (float)cumulative_TGC / cumulative_TC;


/* Display final result */

printf("\n-----\n");

printf("Global ID      : %d\n", global_id);

printf("Number of Semesters   : %d\n", m);

printf("Cumulative Total Credits : %d\n", cumulative_TC);

printf("CGPA                : %.2f\n", CGPA);

printf("-----\n");


return 0;

}
```

TEST CASE TABLE:

TEST CASE	INPUT	EXPECTED OUTPUT	TYPE	RESULT
1	Global ID = 101, semesters = 1, courses = 1, credit = 3, grade = O	SGPA = 10.00, CGPA = 10.00	VALID	PASS
2	Global id = 0	Error – enter a positive Global ID	INVALID	PASS
3	Semesters = 0	Error – enter a positive number of semesters	INVALID	PASS

COMPILATION AND EXECUTION:

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```
(kali㉿kali)-[~/day19]
$ ./sgpa
Enter Global ID: 101
Enter number of semesters: 1

Enter number of courses for Semester 1: 1

Course 1
Enter course code: 123
Enter credit: 3
Enter grade (O/A/B/C/D/E/P): O

Semester 1 SGPA = 10.00

_____
Global ID                : 101
Number of Semesters      : 1
Cumulative Total Credits : 3
CGPA                     : 10.00
_____
```

```
(kali㉿kali)-[~/day19]
$ ./sgpa
Enter Global ID: 0
Invalid Global ID. Enter a positive value: 103
Enter number of semesters: 2

Enter number of courses for Semester 1: 2

Course 1
Enter course code: 345
Enter credit: 4
Enter grade (O/A/B/C/D/E/P): A

Course 2
Enter course code: 678
Enter credit: 3
Enter grade (O/A/B/C/D/E/P): O

Semester 1 SGPA = 9.43

Enter number of courses for Semester 2: 0
Invalid number of courses. Enter a positive value: 2

Course 1
Enter course code: 456
Enter credit: 3
Enter grade (O/A/B/C/D/E/P): A

Course 2
Enter course code: 253
Enter credit: 4
Enter grade (O/A/B/C/D/E/P): B
```

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Semester 2 SGPA = 8.43

Global ID	:	103
Number of Semesters	:	2
Cumulative Total Credits	:	14
CGPA	:	8.93

(kali@kali)-[~/day19]

\$./sgpa

Enter Global ID: 105

Enter number of semesters: 0

Invalid number of semesters. Enter a positive value: 1

Enter number of courses for Semester 1: 1

Course 1

Enter course code: 754

Enter credit: 4

Enter grade (O/A/B/C/D/E/P): 0

Semester 1 SGPA = 10.00

Global ID	:	105
Number of Semesters	:	1
Cumulative Total Credits	:	4
CGPA	:	10.00

.....